



The Great Rebalancing (4/5): The connectivity chokepoint

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- Digital connectivity—the transmission of data through terrestrial fibre, mobile networks, subsea cables and satellites—has become a critical arena of geopolitical competition.
- The Internet's physical infrastructure is increasingly viewed as a source of strategic leverage, resilience and vulnerability.
- In the era of “infrastructure realism,” the contest increasingly involves both states and hyperscalers such as Google, Meta, Microsoft and Amazon, whose AI and cloud ambitions are reshaping global connectivity infrastructure.

Digital connectivity has become a central arena of great-power competition. While states have long used communications infrastructure to project influence and control, in the age of AI, the contest to dominate digital networks has become a defining geopolitical frontier as countries and – increasingly – private actors look to win the AI race.¹

This chapter in our Infrastructure Realism series examines the physical systems that enable digital connectivity: undersea cables and communications satellites. We assess their role on the Internet—the largest and most consequential connectivity network—and the strategic advantages and vulnerabilities they create for state actors.

Subsea cables are indispensable to the global economy. They carry more than 99% of intercontinental data traffic, making sabotage or prolonged disruption a direct threat to global Internet connectivity.² Satellites carry only a small share of international data, but they are crucial for last-mile access where fiber cannot reach, particularly for governments and defense users. As we go on to explore, the rise of communications-based satellites is also becoming integral to the global AI race.

Several incidents have underscored the criticality of connectivity infrastructure. In April, reports of Russian surveillance near UK cables and pipelines raised concerns.³ In May, amid conflict in the Middle East, Iran threatened to sever undersea cables in the Red Sea. Iran's IRGC, also floated the idea, through state-affiliated media, of charging foreign operators infrastructure fees and licensing royalties for routes through Iranian territorial waters.⁴ This would have significant implications for hyperscalers such as Meta, Google and Microsoft, which rely on these corridors for portions of their international traffic and cloud connectivity.

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¹ The first instance of communications sabotage was during WW1, when the British cut Germany's undersea telegraph cables in the English Channel, forcing German telegraph traffic onto British-controlled networks that could be intercepted. And in the 21st century, autocratic states like Iran and Russia have imposed internet shutdowns to control their population / information access.

² The 99% does not account for total global internet traffic, given the vast majority of internet traffic is domestic and does not cross an ocean at all. Instead, the 99% applies to internet traffic that goes between continents. Mauldin, Alan. 2023. 'Do Submarine Cables Account for Over 99% of Intercontinental Data Traffic?'. In *Telegeography.Com*.

³ Olivia Ireland, Thomas Copeland, and Frank Gardner, "UK Says Russia Ran Submarine Operation over Cables and Pipelines," BBC News, April 9, 2026, <https://www.bbc.co.uk/news/articles/cre13qn9z7do>.

⁴ "IRGC Media Float Plan to Levy Fees for Subsea Internet Cables," Amwaj. Media, 2020, <https://amwaj.media/en/media-monitor/irgc-media-floats-plan-to-levy-fees-for-subsea-internet-cables>.



Section 1 explains how the Internet works and traces the physical journey of data, highlighting the points at which state disruption can occur. We give less emphasis to 5G, due to our definition of what constitutes the Internet's infrastructure. This is because while states—most notably China—have used 5G equipment to project influence, 5G is primarily an access network: it links mobile devices to the wider Internet but does not replace the role of subsea cables and satellites.

Sections 2 and 3 analyse subsea cables and communications satellites, detailing the strategic advantages and vulnerabilities of each ecosystem and how states are seeking to consolidate their influence. Finally, Section 4 examines the rise of private actors such as Google and Meta, which are reshaping global connectivity by building their own data corridors and controlling more of the Internet stack to meet rising demand.



1. How connectivity works

1.1 What is the Internet?

The Internet enables global communication, data exchange and access to information through a system of cables, satellites and wireless technologies. It is not a single entity but a common set of protocols that allows independently owned networks to communicate. In this sense, it is a “network of networks” through which data moves across billions of devices.

To move across the Internet, data is broken into small “packets” and routed across different systems using standard protocols. The primary ones are the Transmission Control Protocol (TCP), which ensures the reliability of the connection and data, and the Internet Protocol (IP), which acts as an addressing system to find and route packets to their destination.⁵ Using these protocols, data travels through a global web of fiber-optic cables that connect major network hubs, forming what is known as the Internet Backbone. The process is as follows:

- A packet leaves the user’s device and enters a local access network.⁶ Over fixed broadband, the data packet passes through a Wi-Fi router and then to an Internet service provider (ISP). Over mobile broadband, 5G carries it to a radio base station and the mobile operator’s core network before it reaches the wider Internet.
- From the ISP, the packet is passed from one junction (or router) to the next, crossing multiple networks toward its destination server.
- If the destination server is nearby, the packet may be exchanged at a local Internet Exchange Point (IXP). If it is distant, the ISP routes it onto the backbone, across a country or ocean, and into an ISP at the other end.
- The response takes a similar path back, with each router along the way making forwarding decisions based on the destination IP address.

How one network gets its traffic to the rest of the Internet, and whether it must pay for it, is where key corporations come in. There are two ways networks connect: 1) through transit, where a smaller network pays a larger one to carry its data across the globe, and 2) through peering, where two networks of similar size agree to exchange traffic directly at no cost. Within this system, a hierarchy of companies has emerged.

At the top are Tier 1 network operators, which can reach the entire Internet through settlement-free peering. This small group includes major telecom operators such as Lumen, AT&T, Tata Communications, NTT, Arelion, GTT and Cogent.

Below these are Tier 2 network operators, such as regional carriers and internet service providers like Comcast and Vodafone. These entities primarily purchase transit from Tier 1 providers to reach parts of the internet they cannot access via peering. Finally, Tier 3 network operators rely exclusively on paying for transit from Tier 2s to connect their customers. These are typically the local ISPs that provide Internet service to homes or offices. Therefore, a data packet from a user’s laptop typically travels from a Tier 3 provider to a Tier 2, onto a Tier 1 backbone, across an undersea cable to another Tier 1 network, and then down through the tiers to its destination server.

1.2 How data physically travels

The Internet ultimately depends on physical infrastructure. As we said in step 1 above, the first leg is not always a cable, with mobile technologies like 5G carrying data wirelessly between the user’s device and a nearby base station. Satellite systems can also provide a

⁵ Rus Shuler, “How Does the Internet Work?,” Stanford.Edu, 2002, <https://web.stanford.edu/class/msande91si/www-spr04/readings/week1/InternetWhitepaper.htm>.

⁶ Access networks connect the user’s device or premises to the first point in an Internet or telecom provider’s network. They can be wired or wireless, and in addition to 5G and Wi-Fi mentioned above, other examples are fiber-to-home, cable broadband, and satellite broadband.



wireless connection over much longer distances. Beyond that access layer, however, both systems typically rely on terrestrial fibre and the wider Internet backbone. Most long-distance traffic travels as pulses of light through fibre-optic cables buried on land or laid across the seabed, and each glass strand is thinner than a human hair.

Terrestrial cables handle domestic and regional internet traffic, while undersea fiber-optic cables carry intercontinental internet and telecom traffic. When a user sends data across the globe, it travels through local terrestrial cables to a coastal landing station. This station then transmits the data across the ocean via a subsea cable to a corresponding landing station on another continent, which receives the data and sends it through local terrestrial cables to the destination server.

Satellites complement cables, providing connectivity in remote or disrupted environments where fiber-optic cables cannot reach. There are two different types: traditional satellites, which sit in a fixed, high-altitude orbit, offering broad coverage but with a noticeable delay due to the long signal path. A more recent approach, exemplified by services like Starlink, uses large constellations of thousands of satellites flying much lower to the earth to reduce latency and improve service.

1.3 How the Internet transitioned from a public good to a private advantage

Similar to crypto's arc discussed in this series ([here](#)), the Internet came to symbolise decentralisation during the globalisation and commercialisation of the 1990s and 2000s. Its ability to reroute traffic reinforced the perception that it was neutral and disintermediated. Yet the network depends on physical assets concentrated among a limited number of carriers—and increasingly hyperscalers—that control key digital corridors and possess extensive visibility over network traffic.

Conceived as a “network of networks,” the Internet began as a US government initiative in the 1960s to facilitate information sharing among research universities.⁷ Its predecessor, ARPANET (Advanced Research Projects Agency Network), became the first operational “packet-switching network” in 1969. Famously, its first attempted message – “LOGIN” – crashed the system after only the letters “LO” appeared, an event now widely regarded as the moment the Internet was born. Over the following decades, public researchers cultivated the Internet ecosystem by developing concepts like email, data packet switching, the World Wide Web service (WWW), and the protocols needed to create the Internet ecosystem, all with the goal of keeping the Internet open for all.

By the 1990s, the emergence of major internet exchange providers and the release of WWW software into the public domain enabled millions of people to use the Internet freely.⁸ As private interest in commercializing the Internet grew, the US Congress passed legislation in 1993 that opened access to Internet applications like computing and networking to private enterprise, which had previously only been available to universities and government agencies. This led to thousands of firms entering the market to pursue commercial ventures using TCP/IP technologies.

By 1995, the government agency responsible for overseeing the Internet, NSFNET, was decommissioned and replaced by four privately operated Network Access Points in Washington, New York, Chicago, and California. This marked the Internet's transition from a government-funded academic experiment to a fully private enterprise. The dot-com bubble in the late 1990s and eventual collapse of Internet stocks in the 2000s cleared the way for the companies that today define the modern Internet, including Amazon (1994), eBay (1995), Google (1998) and Facebook (2004). The more recent roll-out of broadband,

⁷ The Department of Defense created ARPA (the Advanced Research Projects Agency) in 1958 in an attempt to lead technological breakthroughs for the military. It was replaced by NSFNET as the main US research network for the Internet in the 1980s.

⁸ The Internet is the global infrastructure of interconnected computers and networks, while the WWW is a service running on that infrastructure, using web browsers to access linked hypertext documents.



fiber-to-the-home, and 3G/4G/5G cellular networks has also helped make the Internet ubiquitous and increasingly mobile.

1.4 Why a decentralized system still creates concentration

No single entity owns the Internet, but a small number of companies and countries control the infrastructure on which it depends on. As a result, the Internet's ecosystem remains highly concentrated across few backbone carriers and transit providers, telecom-equipment suppliers, cable landing stations, and internet exchange points to name a few, thereby creating numerous chokepoints that different countries have raced to ensure a foothold in.⁹

Beyond telecom equipment, the physical layer of digital connectivity is also concentrated. Subsea-cable ownership spans telecom operators, consortia, state-backed entities and technology companies, yet control over capacity, construction, strategic routes and landing points is narrowing. Cable ownership is distinct from Tier 1 backbone status, but the boundary is blurred because operators such as AT&T and Tata also own subsea infrastructure or capacity.

Today, this landscape is also shifting as large content providers have become the dominant users of international bandwidth and increasingly fund, co-own or purchase large allocations of new subsea cable capacity. As we will explain more in section 4, this is flattening the traditional hierarchy, since under these private backbones, hyperscalers are reducing their reliance on traditional Tier 1 transit hierarchy by peering directly with consumer ISPs and even placing their own servers inside those ISP networks.

Figure 1: Notable chokepoints in the connectivity corridor

	Selected key players
Subsea cables (suppliers)	SubCom, Nippon Electric Company (NEC), HMN Technologies, Alcatel Submarine Networks (ASN)
Subsea cables (owners)	Hyperscalers (Google, Meta, Microsoft, Amazon) and Telecom carriers (Orange, Tata, Vodafone, Telstra, China Telecom, China Mobile)
Tier 1 Backbone Operators	Lumen, Arelion, AT&T, Tata Communications, Orange, NTT, GTT and Cogent.
Satellite LEO Constellations	Starlink, OneWeb/Eutelsat, Amazon Kuiper, China SatNet
Telecom Equipment / 5G	Huawei, Ericsson, Nokia, ZTE
Data centres / cloud	AWS, Microsoft Azure, Google Cloud, Alibaba Cloud, Huawei Cloud, Tencent Cloud, Baidu AI Cloud, Oracle Cloud

Source: Deutsche Bank Research, CAIDA.

2. Undersea cables: The invisible backbone of the internet

2.1 How undersea cables work

Subsea cables are long, fiber-optic conduits laid on the seabed that, in total, stretch for more than 1.8 million kilometers. There are approximately 500 active submarine cables around the world, and while many are barely thicker than a garden hose, yet they carry over 99% of all intercontinental data traffic. Their dominance is due to their high capacity, low latency and reliability, which make them ideal for transmitting vast volumes of data and information quickly. While most of the world's cables are laid by just four companies – SubCom (US), Alcatel Submarine Networks (France), Nippon Electric Company (Japan),

⁹The geopolitical case of China's Huawei is one example. As the company that provided many of the physical and software components needed for 5G, Huawei rapidly ascended to become a significant part of the Internet's underlying infrastructure, and thus a critical player in global digital connectivity. This later proved to be a problem during accelerating tensions between China and Western countries, as many nations sought to disentangle themselves from Huawei's supply chains in their telecom networks.

and HMN Technologies (formerly Huawei Marine Networks; China) – it is US hyperscalers who now finance and control most of the new cable capacity.¹⁰

Today, subsea cables are the foundation of the global telecommunications industry because they transfer data and voice traffic more cheaply and quickly than any alternative. Yet for much of the 20th century, the bulk of transoceanic traffic was primarily carried by satellites. The transition only began in 1988 with the installation of TAT-8, the first trans-oceanic fibre-optic cable, which linked the US, the UK and France. This marked the point at which submarine cables began to outperform satellites in volume, speed and cost-effectiveness of data and voice communications.¹¹ Consequently, almost all transoceanic telecommunications are now routed via the submarine cable network, with satellites handling only around 3% of total traffic now, according to ESCAEU.¹²

Figure 2: Global submarine cables, with the most geopolitically contentious corridors



Source: TeleGeography, www.submarinecablemap.com, Deutsche Bank Research.

2.2 Why undersea cables are strategic

Beyond serving as the Internet's backbone, these underwater cables are essential for global trade, finance, cloud computing, and government communications. For example, submarine cables can carry around 375mn simultaneous telephone conversations on 1 fiber optic strand.¹³ The SWIFT network also relies on cables for transmitting financial data to more than 8,300 member financial institutions in 195 countries.¹⁴ It is also estimated that \$10trn worth of financial transactions travel through these subsea cables every day.¹⁵

A 2026 study found that undersea cable disruption also had massive macroeconomic consequences for global economies.¹⁶ Specifically, a typical cable disruption often triggers an immediate 2ppt decline in GDP capita growth for an economy. There is also evidence to show that the impact continues to accumulate over time; within three to six

¹⁰ The number of systems by supplier is ranked by ASN (23), Subom (9), NEC (8), and HMN Tech (6). Data taken between 2021 – 2025. Kieran Clark, "Submarine Telecoms Industry Report," SubTel Forum, October 30, 2025, <https://subtelforum.com/submarine-telecoms-industry-report/>.

¹¹ Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.

¹² "Submarine Telecommunication Cables." n.d. www.Escaeu.Org. Accessed July 7, 2026. <https://www.escaeu.org/articles/submarine-telecommunications-cables/>.

¹³ Submarine Telecommunication Cables." n.d. www.Escaeu.Org. Accessed July 7, 2026. <https://www.escaeu.org/articles/submarine-telecommunications-cables/>.

¹⁴ Swift data

¹⁵ "Securing Europe's Subsea Data Cables." 2024. Carnegie Endowment for International Peace. 2024. <https://carnegieendowment.org/research/2024/12/securing-europes-subsea-data-cables?lang=en>.

¹⁶ Joël Cariolle. Digital Disasters: The Macroeconomic Costs of Submarine Cable Breaks. 2026. (hal-05544990)



years, the affected country's economy diverges, falling 9 ppts below its non-disrupted neighbors. These disruptions also reduce overall economic efficiency, innovation and labor productivity, followed by a tightening of financial conditions as domestic credit for non-financial private agents' contracts and cross-border banking positions weaken. The study also noted that in countries with fewer international cable relations and/or without internet exchange points, disruptions widen the GDP per capita gap immediately and substantially over the medium run, compared to nearby non-disrupted economies that instead experience offsetting gains consistent with reallocations.¹⁷

These findings highlight a broader point: the resilience of subsea cable infrastructure is not merely a telecommunications issue, but a broader precondition for economic stability and the functioning of modern financial systems.

The strategic importance of these networks is further amplified by the geography of the routes themselves. Because international data traffic is concentrated within a limited number of physical corridors, damage to key cable systems could have magnanimous consequences. One example that illustrates this point is to consider a high-frequency trader in London pulling live market data from a server on Wall Street in New York. The request would be transmitted as packets across terrestrial fibre networks in the UK, where it enters a low-latency transatlantic cable system. Examples of such routes include EXA Infrastructure's Express system, which forms part of the low-latency route between New York and London or Aqua Comm's AEC-1 linking New York and Ireland, and then on to the UK.

Many of these transatlantic cables pass through the GIUK Gap in the Atlantic, a geopolitically sensitive maritime corridor between Greenland, Iceland, and the UK. The passage is contentious because it is a critical chokepoint for NATO maritime surveillance and because it is also increasingly a primary route for Russia's Northern Fleet to access the Atlantic. So this concentration of critical data infrastructure within a militarily significant area creates a potential vulnerability.

As such, subsea cable infrastructure security has become a national priority for several countries, including the EU, US, Japan, India and Australia.

2.3 Undersea cable vulnerabilities

The criticality of subsea cables is heightened by the fact that they can be vulnerable to damage, outages and cyber risks, such as natural or man-made hazards, and hazards to other infrastructure including the landing stations and IT systems that support the cables. This systemic risk is amplified because many global networks rely on a limited number of routers, hubs and suppliers.

Crucially, there is often no "Plan B" when a major cable is disrupted, and there are a reported 150-200 cable faults recorded annually.¹⁸ The challenge is threefold: first, repairs are immensely costly, estimated at \$500,000-\$1mn per incident for a subsea fibre optic cable, and \$10 - \$100mn per incident for a subsea power cable.¹⁹ Second, there are only ~40 specialist cable repair vessels in the entire world, owned by a handful of companies like SubCom, Orange Marine, and HMN Technologies. These vessels operate on a first come first served basis within the consortium's zone agreement. And while the repair ships can mobilize within a 24-hour period, repair time frames vary according to fault location, while the requirement to secure repair permits within territorial or exclusive economic zone waters can be burdensome and significantly delay starting operations.²⁰ Finally, these cables are often concentrated in physical chokepoints – such as along

¹⁷ Joël Cariolle. Digital Disasters: The Macroeconomic Costs of Submarine Cable Breaks. 2026. <hal-05544990>

¹⁸ International Cable Protection Committee.

¹⁹ Qiu, Winston. 2025. "Statistics on Subsea Cable Fault and Repair - Submarine Networks." Submarinenetworks.Com. June 28, 2025. <https://www.submarinenetworks.com/en/nv/insights/statistics-on-subsea-cable-fault-and-repair>.

²⁰ Constable, Mike, Burdette, Lane, Mauldin, Alan. "The Future of Submarine Cable Maintenance: Trends Challenges, and Strategies." June 2025.



shallow straits or continental shelves – that are also areas of geopolitical strife, making conducting any reparations difficult in this environment.

While direct evidence of deliberate cable sabotage has been hard to prove thus far, several incidents of damage to critical undersea infrastructure sites have caught the attention of governments.²¹ The US Congress, in its proposed Strategic Subsea Cables Act of 2026, noted damage to cables connecting the Norwegian mainland with the Norwegian archipelago of Svalbard (2022), multiple cuts to cables in the Taiwan Strait (2023, 2025), and disruptions to cables in the Baltic Sea (2024, 2026) as key examples of this growing threat.²²

Perhaps the more immediate concern for individual users was the threat that emerged earlier this year, when Iran threatened to cut its undersea cables in the Red Sea. Since these routes carry around 17-25% of global internet traffic, the ramifications of such an act would be severe.²³ Even before this threat, subsea cable operators in the Red Sea have been plagued with numerous problems. Yemeni territorial disputes and Houthi attacks on shipping lanes have obstructed the deployment of several new cables – including the 2Africa cable, which is almost complete, and other cables such as the IEX, Africa-1, and Raman, which are still awaiting deployment. The geopolitical strife in the corridor has led to recent reports that all cable projects there have been “delayed indefinitely.”²⁴

Geopolitical risk is rising, but most cable damage remains accidental. The International Cable Protection Committee estimates that fishing gear and ship anchors account for roughly 70–80% of faults.

2.4 State actors responding

Governments have long regarded submarine telecommunications networks as critical infrastructure requiring strong protection (for example, ACMA, 2007).²⁵ More recently, they have also recognized the geopolitical value of controlling these corridors. In 2015, China set a goal of capturing 60% of the fibre-optic cable market through state-backed firms under its Digital Silk Road initiative.²⁶ HMN Technologies (formerly Huawei Marine) is one of four companies that collectively dominate global cable manufacturing.

However, China’s influence is primarily concentrated in the Global South, as Chinese firms have been largely excluded from most Western-connected systems since the US launched its 2020 Clean Network program to counter China’s influence across telecoms and data. Under the Clean Network, Chinese telecom and cable firms were shut out of lucrative bids for cable-laying projects, most notably when HMN Technologies lost the SEA-ME-WE 6 project (running from France to Singapore via the Suez) in 2023 despite its competitive pricing. aChina has pivoted to financing new cable projects connecting Africa, Asia and the Middle East, such as its flagship PEACE cable (Pakistan-East Africa-Europe) and EMA (Europe-Middle East-Asia) project, which is backed by three state-owned telcos: China Telecom, China Mobile, and China Unicom in partnership with Egypt and Saudi Arabia.

Over the last few years, the US has consolidated its toolkit to protect its subsea cable security. In February of this year, the US Congress introduced the bipartisan Strategic

²¹ In 2025 Western media discovered that engineers at China’s Lishui University in Zhejiang had filed a patent for the creation of a submarine cable cutting device back in 2020. While it was argued it was needed for emergency situations, the announcement raised suspicion amongst Western media, which uncovered the patent applications in January 2025.²² And in April 2026, the South China Morning Post reported China had successfully tested cable cutting 11,483 feet (3,500m) undersea.

²² Joe Wilson. 2026. US Strategic Subsea Cables Act. https://joewilson.house.gov/sites/evo-subsites/joewilson.house.gov/files/evo-media-document/undersea-cables-signed_0.pdf.

²³ [Iran Signals a New Threat to the Gulf’s Internet | WIRED Middle East](#)

²⁴ “Stalled Subsea Cable Projects Threaten Middle East Digital Ambitions.” 2026. FinancialTimes. Financial Times. June 25, 2026. <https://www.ft.com/content/5e6ce736-3a48-406a-933a-ed215370bd16?syn-25a6b1a6=1>.

²⁵ Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.

²⁶ “The Digital Silk Road: A Growing Priority for Beijing as Its Tech Champions Expand Overseas.” 2024. Merics. March 27, 2024. <https://merics.org/en/tracker/digital-silk-road-growing-priority-beijing-its-tech-champions-expand-overseas>.



Subsea Cables Act. A core proposal of the act requires the President to impose sanctions on foreign actors who intentionally damage critical undersea infrastructure, including subsea fiber optic cables.²⁷ If passed, this act will build upon the US's 2020 Undersea Cable Control Act, which focused more on preventing adversaries from accessing technology to build and operate undersea cables. Together, the two bills aim to deny countries like Russia and China the ability to conduct what a sponsoring senator referred to as “shadow warfare” against US critical infrastructure in subsea cables.²⁸

Finally, a landmark development to watch is the recent US Federal Communications Commission (FCC) vote to extend cable surveillance to terrestrial endpoints, such as cable landing stations and the hardware housed within them that converts the subsea optical fiber signals. The FCC has identified these endpoints as chokepoints vulnerable to data interception, physical sabotage, and cyber espionage. Under the new rules, hyperscalers and tech firms employing AI infrastructure could receive expedited licensing for their terrestrial hardware, while foreign rivals would need to apply for individual licenses and face the possibility of denial. This is a potential win for the hyperscalers, although the pool of eligible hardware suppliers may significantly shrink, creating short-term bottlenecks in cable landing equipment manufacturing.²⁹

Europe has historically had a strong presence in the subsea cable industry, but this position is being challenged by competing US and Chinese interests and the rise of US hyperscaler control. France remains a leading global supplier through its ownership of Alcatel Submarine Networks and Orange Marine, while Italy's Elletra holds a strong mid-tier position.

The EU is attempting to carry out a coordinated approach towards cable security via its 2025 EU Action Plan on Cable Security, but implementation has been slow. In June of this year, however, the European Commission finalized a €5.8mn grant to establish the first two regional cable hubs in the Baltic Sea and the Mediterranean Sea, and a separate €40mn fund to increase its submarine cable repair capacity.³⁰

Finally, US-allied attempts to find a geopolitically secure “Fourth Route” away from volatile transits in the Red Sea and South China Sea have led to the creation of the Far North Fiber express route. This will be the first long-haul submarine cable connecting Europe and Asia exclusively through Allied-sovereign waters. It is expected to cost \$1.17bn and should be ready for service by the end of 2026.³¹ The project is owned by US Alaskan company Far North Digital, in conjunction with its Canadian affiliate True North Global Networks, but it has since expanded into a consortium with companies from Japan and Finland as well. Alcatel Submarine Networks will supply the cables.

All of this strategic interest has translated into unprecedented subsea cable infrastructure investment, estimated to have averaged over \$2bn per year in new construction over the last 9 years.³² Most new investment is forecast to be directed toward trans-pacific routes.

The most consequential response, however, is coming from the private sector, which controls most of the infrastructure. AI is sharply increasing bandwidth demand and prompting content and cloud providers—notably Google through its Pacific Connect

²⁷ Qiu, Winston. 2026. “US Introduces Strategic Subsea Cables Act of 2026 — Senior Industry Analyst Briefing.” Submarine Networks. April 14, 2026. <https://www.submarinenetworks.com/en/nv/insights/us-introduces-strategic-subsea-cables-act-of-2026>.

²⁸ “Wilson and Meeks Introduce ‘Strategic Subsea Cables Act Of 2026.’” 2026. Representative Joe Wilson. March 27, 2026. <https://joewilson.house.gov/media/press-releases/wilson-and-meeks-introduce-strategic-subsea-cables-act-2026>.

²⁹ Qiu, Winston. 2026. “FCC Overhauls Subsea Cable Licensing: SLTE Licensing and Fast-Track Process for US Tech Giants.” Submarine Networks. June 26, 2026. <https://www.submarinenetworks.com/en/nv/insights/fcc-subsea-cable-licensing-slte-fast-track-process>.

³⁰ Ernst, Lauren. 2026. “EU Launches Regional Cable Hubs To Protect Critical Undersea Infrastructure.” Ocean News & Technology. June 23, 2026. <https://oceannews.com/news/subsea-cable/eu-launches-regional-cable-hubs-to-protect-critical-undersea-infrastructure/>.

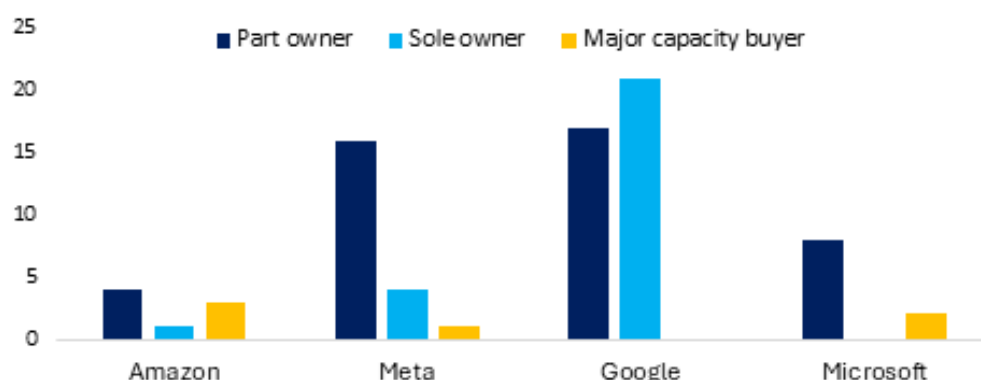
³¹ “Far North Fiber - Submarine Networks.” 2026. Submarinenetworks.Com. 2026. <https://www.submarinenetworks.com/en/systems/trans-arctic/far-north-fiber>.

³² Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.



Initiative and Meta through Project Waterworth—to accelerate investment.³³ This private dominance means resilience will depend as much on international coordination with corporate owners as on national policy.

Figure 3: Number of submarine cables owned by content providers



Source: Telegeography, Deutsche Bank Research..

3. Satellites: Beyond terrestrial networks

Like subsea cables, satellites form a part of the Internet's physical layer. Throughout much of the 20th century, satellites carried the bulk of all transoceanic digital connectivity. The launch of Telstar, the first operational telecommunications satellite in 1962, was soon followed by other satellites that helped commercialize the international communications and broadcast industry.

Today, however, traditional satellite communications (satcom) complement rather than replace cables. Satellites provide broadcast and communication services to sparsely populated regions not served by terrestrial infrastructure and act as a strategic back-up for disaster-prone regions. While subsea cables deliver more consistent, high-capacity links between major population centres, satellites are a cost-effective solution for governments seeking to connect regions with low population densities.³⁴

The renewed focus on satellite communications is driven less by their current share of global traffic than by their strategic potential. Two forces are decisive. First, the war in Ukraine and US-China technology competition are prompting sovereign investment in space-based communications. Second, rising AI workloads are intensifying demand for bandwidth and encouraging the deployment of satellite mega-constellations. We examine the implications in greater detail in the next chapter.

For these reasons, communications-based satellite networks and space-based connectivity represent the next major frontier in the contest for control over digital connectivity.

3.1 How satellites support connectivity

There are three main types of satellites in operation: communications, Earth-observation, and radiofrequency- based. This section focuses on communication satellites, which provide services ranging from radio, TV, telecommunications, and Internet by receiving signals from Earth and retransmitting them to a ground station elsewhere, creating a communications channel between distant regions that were previously unable to communicate with one another.

³³ The State of the Network: 2026 Edition. 2026. Telegeography.

³⁴ Carter L., Burnett D., Drew S., Marle G., Hagadorn L., Bartlett-McNeil D., and Irvine N. (2009). Submarine Cables and the Oceans – Connecting the World. UNEP-WCMC Biodiversity Series No. 31. ICPC/UNEP/UNEP-WCMC.



Traditional satellite communications (also referred to as “satcom”) rely on large geostationary (GEO) satellites positioned about 35,800 kilometres above Earth. Systems such as the early Intelsat 1 (1965) and the more recent Jupiter 3 (2023), provide broad, fixed coverage with high availability and favourable ground-terminal geometry.³⁵ GEO assets have historically supported predictable business models because of their long operating lives, typically around 15 years, and high launch costs. Their strategic value is clear, but so is their limitation: distance creates latency. The farther a satellite is from Earth, the longer data takes to travel, which is one reason subsea cables continue to carry most global connectivity.

In addition, GEO satellites also suffer from increasingly longer development cycles and higher integration complexity. As a result, both state and private actors are turning their interest towards low-earth orbit (LEO) satellites instead. LEO satellites orbit between 160 and 1,500km above the Earth’s surface, enabling a more seamless user experience for applications like live streaming, video calls, remote desktops, etc. Because they circle the planet up to 16 times a day, LEO constellations can provide ubiquitous coverage to nearly any point on Earth. SpaceX’s Starlink was the harbinger of the LEO satellite model, but several LEO constellations have emerged in response to Starlink since then, including Amazon LEO, OneWeb (now part of Eutelsat), Telesat Lightspeed, and China’s Guowang and Qianfan projects. In fact, in July 2026 Amazon announced it had filed an application with the US FCC to deploy a constellation of more than 5,100 satellites that would provide direct-to-device connectivity by 2028.³⁶ Doing so will directly challenge Starlink Mobile, a subset of Starlink comprised of around 650 satellites that beam mobile data to phones in remote regions on earth, and which has recently also announced plans to launch “thousands” more satellites by end of 2028.³⁷

3.2 Why satellites are strategic: the arrival Starlink

Although subsea cables carry the bulk of internet traffic, communications-based satellites are also integral to the global economy. They provide connectivity to remote and underserved regions that undersea cables cannot reach, while also enabling communications for mobile platforms such as ships, aircraft and military assets. Indeed, industries such as aviation, rail, shipping and logistics depend on satellites for navigation, route optimization, and cargo tracking.

Perhaps to an even greater extent than cables, satcom has become strategic to a country’s military infrastructure. Satcom companies are developing partnerships to cater to civilian and government forces. For example, Canada’s Telesat (c. 1969) announced a strategic partnership with the Canadian Armed Forces to develop and deliver a multifrequency, Arctic military satellite communications capability via its forthcoming Lightspeed constellation, which the company targets to launch in late 2026.³⁸ Meanwhile, established satcom companies with significant government -related revenue streams, like Viasat, SES and Eutelsat, have seen strong growth over the past year. Our DB equity analysts estimate that existing GEO communications contracts alone account for around \$50bn-60bn in defence-related spending (excluding non-US allies).³⁹

The greatest harbinger has been SpaceX’s Starlink, which has provided encrypted communications, imagery and geolocation services for the Ukrainian military against Russia since 2022. In addition, Starlink’s “direct-to-device” business model that connects smartphones directly to satellites threatens to collapse the traditional, hierarchical

³⁵ Yu, Edison, Roshan Ranjit, Scott Deuschle, Christophe Menard, and Laura Li. 2026. “State of the Private Markets: Rise of Micro-GEO Satellites.” Deutsche Bank Research.

³⁶ “Amazon Targets Musk’s Starlink with Satellite Constellation for Mobile Services.” 2026. FinancialTimes. Financial Times. July 27, 2026. <https://www.ft.com/content/83e93ce0-1ebd-4b04-a70c-82ff7b64a5f4?syn-25a6b1a6=1>.

³⁷ Kan, Michael. 2026. “Starlink Mobile Will Span ‘Thousands’ of Satellites By 2028.” PCMag UK. August 3, 2026. <https://uk.pcmag.com/networking/166530/starlink-mobile-will-span-thousands-of-satellites-by-2028>.

³⁸ Yu, Edison, and Laura Li. 2026. “Telesat: Focus Increasingly Shifts toward Defense.” Deutsche Bank Research.

³⁹ Yu, Edison, Roshan Ranjit, Scott Deuschle, Christophe Menard, and Laura Li. 2026. “State of the Private Markets: Rise of Micro-GEO Satellites.” Deutsche Bank Research.



Internet infrastructure, where cables traverse terrestrial data and comms-based satellites take it on its last leg.

Starlink's ability to bypass the conventional hierarchy of connectivity provided vital communications for Ukrainian forces and civilians at a time when conventional communications like fiber lines, cellular towers and power grids were damaged early in the war. This highlighted how a private company could become essential to a nation's military capability during a war. The gravity of this power was demonstrated again in February 2026, when SpaceX and Ukraine introduced a terminal verification system that effectively blocked Russia from using Starlink military terminals, after it was revealed that Russia had acquired Starlink terminals via third-party countries and grey markets to conduct strike drones.

The entry of Starlink in 2019 has shifted the goalposts and urgency for more communications-based satellites. Bringing affordable, high-speed internet to those that lack access to traditional broadband, Starlink's true purpose is to help deploy Musk's goal of AI compute at a massive scale. In fact, Musk confirmed that Starlink is also designed to support a project called Starwind, which aims to place massive data centers in space. He recently filed applications with the FCC for a constellation of space-based AI data center satellites through this project. These space AI data centers will compute in space but then use lasers to offload the processed data directly into the Starlink constellation, which will beam the results back to Earth with low latency.

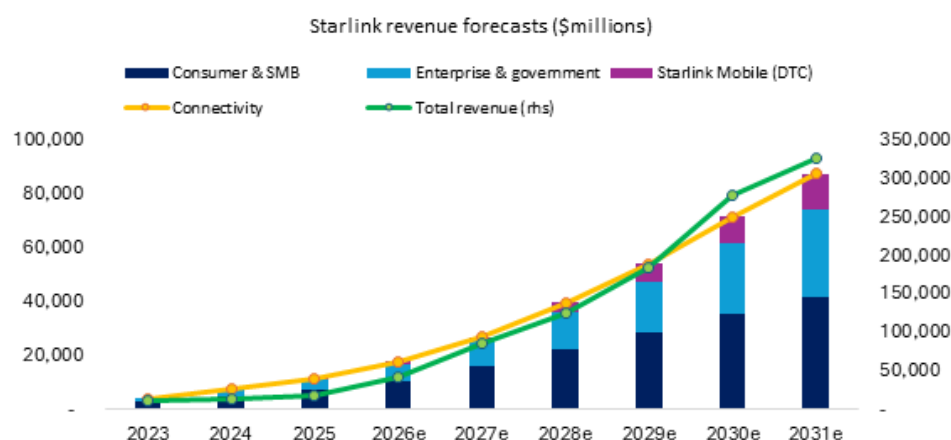
By using Starlink to create a reliable and robust orbital network used for Earth, Musk hopes to eventually extend the satellite internet service to the Moon, Mars, and beyond. Our equity analysts place Musk's aims for orbital data centers within a 5-year horizon and human voyage and colonization in Mars and off-world within 10-year horizon.

Starlink has upped the game for the rest of the world considerably. First, by its sheer dominance in the number of satellites it operates and in the numbers of subscribers it now has on its network. Starlink now accounts for 54% of all operational satellites in orbit,⁴⁰ and our equity analysts estimate Starlink's consumer broadband subscribers will grow to around 17mn by the end of 2026.⁴¹ Second, its ability to simultaneously increase speeds while cutting hardware prices has made it very hard for anyone else to compete. Finally, Starlink's entry into LEO also made it harder for pure-play GEO industry players like Viasat and Intelsat to focus solely on consumer broadband, although as mentioned above there is a growing market opportunity for these players to enter the government services market. We will go more in depth on Starlink and its parent company SpaceX in our final chapter on Cloud and Data Centers.

⁴⁰ "10,400+ Starlink Satellites in Orbit — Live Count (2026) | Orbital Radar." 2026. Orbital Radar. May 31, 2026. <https://orbitalradar.com/how-many-starlink-satellites>.

⁴¹ Yu, Edison, Bryan Kraft, Ross Seymore, Benjamin Black, Laura Li. 2026. "SpaceX: Apex of Civilization Ambition." Deutsche Bank Research.

Figure 4: Starlink connectivity revenue to reach \$18bn in 2026 and \$88bn by 2031



Source: Deutsche Bank Research equity research, company filings.

3.3 Satellite vulnerabilities

Because communications-based satellites are so critical, their vulnerability to deliberate attacks is a matter of strategic concern. Satellites are susceptible to attacks such as jamming (when the uplink or downlink frequency is filled with noise so the signal cannot get through) and spoofing (transmitting fake signals) of their radio waves. Countries like Finland, Estonia and Poland have all reported [GNSS](#) disruptions associated with alleged Russian activities in the Baltic region, which has affected numerous services such as commercial aircraft and maritime navigation. Other examples of disruption include cyberattacks on satellites via their ground stations and control networks, as well as physical and cyberattacks on the orbiting satellites themselves. Notable incidents include evidence that Russia was conducting widespread [jamming](#) of Starlink GPS signals soon after their invasion of Ukraine in 2022. In 2021, Russia also conducted an anti-satellite weapon test that destroyed an inactive satellite and generated a massive field of orbital debris.

Figure 5: Major satellite networks in orbit

Constellation	First launched	In orbit	Total planned
Starlink (US)	May 2019	10,877	42,000
Amazon Leo (US)	Oct 2023 (prototype satellites); production deployment began Apr 2025, initial commercial service expected in late 2026	390+	7,700
OneWeb (UK/Eutelsat)	Feb 2019, merged with Eutelsat in 2023.	654	Gen-1 complete, additional 440 satellites planned with Airbus
Guowang (China)	December 2024	177	12,992 (~13,000) filed with the ITU
Qianfan (China)	August 2024	238	Over 15,000 satellites approved for the final system

Source: Satellitemap.space, ChinainSpace, CNBC.

3.4 How state actors are responding

A rapid, high-stakes military space race looks to be underway between the US, China, and Russia, focused on establishing, securing, and defending satellite communication (satcom) constellations, particularly those in Low Earth Orbit (LEO).

While US companies currently dominate the satcom landscape, they face a growing challenge from other countries, especially China. China is developing its own satcom



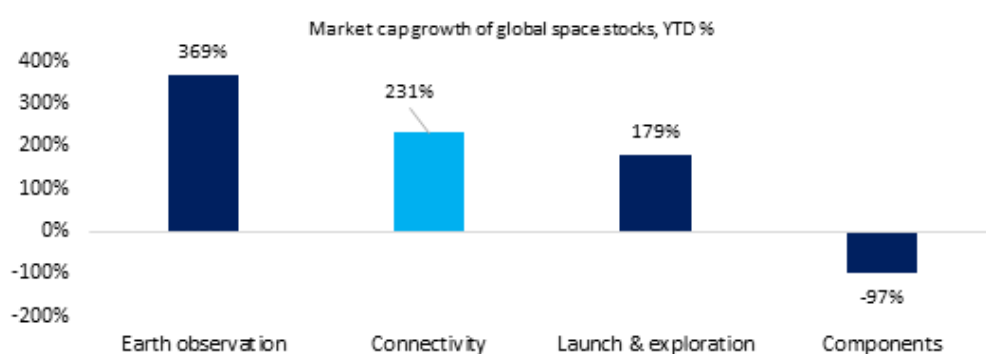
ecosystem through two key initiatives: the state-run Guowang network, aimed towards China's domestic telecom and national security, and the Shanghai-backed Qianfen constellation, which is aimed at the commercial and export markets. Together, these projects aim to deploy between 13,000 and 15,000 satellites.⁴² Guowang is designed primarily to connect rural areas and government institutions, while Qianfan is more commercially focused, aiming to provide in-flight connectivity (IFC) and direct-to-device (D2D) communications, with active pilots underway in Brazil, Malaysia, and Thailand.

Both constellations are in their early stages, with fewer than 200 satellites each currently in orbit. In December 2025, China also submitted filings with the International Telecommunication Union (ITU) to reserve orbital slots and spectrum for nearly 200,000 additional satellites – a move to secure future resources, even though China does not yet have the launch capacity to deploy such a number.⁴³ To facilitate more launch activity, China has also implemented new regulatory guidelines to facilitate IPOs for firms developing reusable rockets.

The EU is also attempting to reduce its dependence on US private operators by building IRIS, a government-controlled constellation of approximately 290 satellites, with services targeted to begin in 2029. However, progress on the project remains slow. Other notable sovereign interests include the UK's stake in OneWeb and Canada's investment in Telesat's Lightspeed project.

Driven by these strategic imperatives, the market for communications-based satellites is growing, as they become critical infrastructure for consumers, enterprises, and governments. Deutsche Bank equity analysts estimate the satcom market grew by over 35% YoY to nearly \$26bn in 2025, mostly driven by consumer broadband and government contracts.⁴⁴ This growth is expected to continue, with market revenue projected to reach \$65-70bn by 2030.

Figure 6: Oversized demand for comms-based satellites has led to a rally in space connectivity stocks



Source: Bloomberg Terminal, Deutsche Bank Research. Returns up to August 2026. Connectivity stocks: AST SpaceMobile, APT, Beijing BDStar, China Satcom, Echostar, Eutelsat, Globalstar, Gogo, Iridium, JSAT, SES, Space42, Telesat, Thaicom, Viasat.

⁴² Yu, Edison, Bryan Kraft, Roshan Ranjit, Lee Horowitz, and Laura Li. 2026. "Space & Satellites: Market Growing and Becoming Critical Infrastructure." Deutsche Bank Research.

⁴³ "China ITU Filing to Put ~200K Satellites in Low Earth Orbit While FCC Authorizes 7.5K Additional Starlink LEO Satellites – IEEE ComSoc Technology Blog." 2026. Comsoc.Org. January 13, 2026. <https://techblog.comsoc.org/2026/01/13/china-itu-filing-to-put-200k-satellites-in-orbit-fcc-authorizes-7-5k-additional-starlink-leo-satellites/>.

⁴⁴ Yu, Edison, Bryan Kraft, Roshan Ranjit, Lee Horowitz, and Laura Li. 2026. "Space & Satellites: Market Growing and Becoming Critical Infrastructure." Deutsche Bank Research.



4. The new power map of connectivity

4.1 The rise of private control

While cloud and space data centers will be the focus of our next chapter, this section zooms in how connectivity supply chains are being affected by the AI build-out.

In the connectivity model set out in Section 1, hyperscalers own the destination of data: the cloud, where applications, websites and AI models run. Historically, these companies have relied on telecom operators for transmission and concentrated investment on data centers and cloud services. That division is now eroding, as hyperscalers are increasingly acquiring, funding or building the underlying infrastructure, including subsea cables and communications satellites—that carry data to and from their platforms.

AI's data intensity is accelerating this vertical integration. In 2016, traditional backbone providers carried most international bandwidth; today, content and cloud networks account for nearly three-quarters of demand.⁴⁵ Owning connectivity allows hyperscalers to lower long-run unit costs, scale capacity with cloud demand and reduce dependence on third-party networks. It also gives them greater control over the economics and governance of the data chain.

This trend is evident in new cable construction, where private companies are starting to swallow up market share. For example, of the \$14bn projected investment in new submarine cables entering service between 2025 and 2027, driven largely by content providers such as Google and Meta rather than the traditional internet backbone providers.⁴⁶ While traditional telecom operators still account for the majority of existing builds, this new investment demonstrates the steady and growing influence of large-scale cloud and content companies in shaping global connectivity.

The same dynamic is beginning to emerge in the satellite ecosystem, where hyperscalers and space technology companies are exploring space-based data centers as a potential long-term solution to the terrestrial constraints that increasingly accompany the buildout of AI compute capacity.

To make orbital computing viable, companies need communications infrastructure capable of moving massive amounts of data between space and Earth, creating a strategic rationale for greater ownership of satellite networks and related connectivity assets. A notable entrant is Jeff Bezos' Blue Origin, which announced TeraWave in January 2026, a multi-orbit communications network comprising 5,280 satellites in LEO and 128 satellites in MEO. Blue Origin plans to begin deploying the constellation in 4Q27 and is targeting enterprise, data-center, and government customers. DB analysts estimate that the constellation could require ~\$18bn of capex.

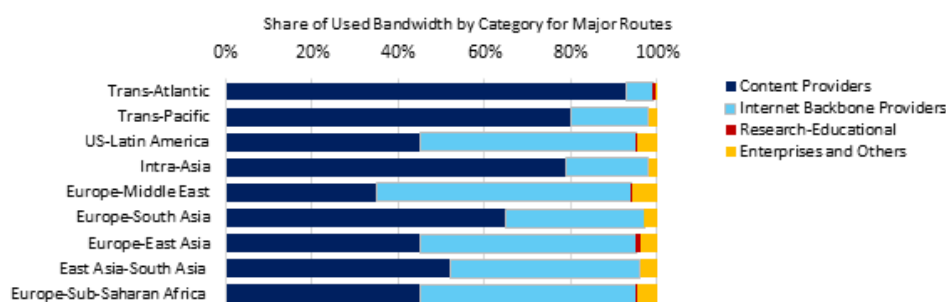
Although Blue Origin has not explicitly positioned TeraWave as infrastructure for orbital data centers, its high-capacity, optically interconnected network could eventually provide part of the communications layer required to connect space-based compute systems with terrestrial networks. In addition to SpaceX and Blue Origin, Amazon is building its LEO communications constellation, while Nvidia-backed Starcloud has filed for regulatory approval to deploy orbital data-center satellites. Starcloud also plans to launch AWS Outposts hardware aboard its second satellite in October 2026, providing an early test of AWS infrastructure operating in space.

⁴⁵ The State of the Network: 2026 Edition. 2026. Telegeography.

⁴⁶ Ibid.



Figure 7: Content networks represent around three-quarters of all global bandwidth demand



Source: Deutsche Bank Research. Data inferred from Telegraphy's 2026 State of the Network Report. Represents used bandwidth as of year-end 2024.

4.2 Effects of the AI disruption

The result is a sharp shift in control over global connectivity. Traditional carriers are increasingly becoming minority consortium members or implementation partners, while hyperscalers lead funding and technical design. The Internet's decentralised architecture increasingly coexists with concentrated private ownership of critical connectivity infrastructure.

The consequences are substantial. By owning subsea cables and satellite networks, hyperscalers are increasingly able to build private connectivity corridors linking their data centres, reducing dependence on traditional telecom operators. This vertical integration bypasses—and may ultimately compress—the tiered, multi-provider structure that has defined telecommunications for decades.

This disruption has forced traditional telecom operators to either merge to survive or retreat into more niche services. By shrinking the telecommunications market, the remaining players can better afford the massive capex required for increasing broadband demand. Notable deals in recent years include \$7.3bn merger between US-based Viasat and British satellite firm Inmarsat in 2023, SES's acquisition of Intelsat for \$3.1bn in 2024, and the \$8bn Rocket Lab/Iridium⁴⁷ consolidation in 2026. By creating these multi-orbit groups, they have been able to concentrate on key segments of mass-market connectivity.

Europe is perhaps at the greatest disadvantage in this new connectivity map, finding itself crowded out by US-led private innovators and state-backed Chinese companies. The European Centre for Development Policy Management argues that the growing concentration of subsea-cable ownership and operational control among US hyperscalers is creating structural dependencies that could erode Europe's digital sovereignty.⁴⁸ This concern is particularly acute for Europe, where ownership and operational control of critical subsea infrastructure is increasingly concentrated among non-EU actors. Unlike the United States, which exercises oversight through institutions such as the FCC, Europe lacks a unified governance framework for the subsea cable ecosystem, raising concerns about long-term digital sovereignty.⁴⁹

The same risk exists in space, where Starlink dominates the global satellite network. As the CEO of the French telecoms operator, Bouygues Telecom, warned in an interview earlier this year, Europe needs to "get some sovereignty" or else it risks having its

⁴⁷ Deutsche Bank and/or its affiliates is acting as advisor to Rocket Lab Corporation on the potential acquisition of Iridium Communications, Inc..

⁴⁸ Pearson, Sasha. n.d. "Troubled Waters: Europe's Subsea Telecommunications Network." Ecdpm. <https://ecdpm.org/work/troubled-waters-europes-subsea-telecommunications-network>.

⁴⁹ Ibid.



connectivity controlled by a non-state actor like Starlink.⁵⁰ Reflecting these pressures, in June 2026, Bouygues itself partnered with Orange and Free-illiad Group to acquire and break up French telecommunications company SFR for €20.35bn.

In the space arena, the EU has attempted to address this problem with its proposed Space Act, which aims to cultivate European champions. However, this has created friction with the US, whose FCC has publicly warned that Washington would retaliate if Europe imposed protectionist measures that favored its own satellite companies over American ones.⁵¹ In the meantime, the formidable presence of Starlink continues to drive a new wave of private communications-based satellite initiatives seeking to compete.⁵²

States therefore face a strategic dilemma. AI and cloud adoption are directing more traffic through hyperscaler-owned networks, increasing dependence on private technology companies even as governments must guarantee resilience and control in a crisis. The US is best positioned because its firms dominate the industry. European governments have responded by buying stakes in, or taking control of, strategic telecom assets, including Italy's acquisition of TIM Sparkle and Spain's 10% stake in Telefónica.

Figure 8: Aside from 2022, where activity dropped sharply amidst the backdrop of Russia Ukraine, the telecoms M&A industry has seen massive growth



Source: Deutsche Bank Research, Dealogic. Data only draws upon publicly available deal value at announcement data. Data up to July 2026.

5. Conclusion

Digital connectivity emerged from an ideal of neutrality and disintermediation. The Internet was designed to share knowledge across research institutions, while cables and communications satellites expanded international exchange. Today, connectivity is an arena of state and corporate power. Control over data flows is becoming as consequential as control over payment infrastructure, and the race for AI leadership is intensifying the strategic value of the networks that carry information.

Subsea cables are uniquely fragile because their criticality is matched by limited substitutability. The parallel with payments is instructive: excluding a country from SWIFT can amount to financial siege; severing its cable links can impose a digital one. Yet payments have emerging alternatives—including CIPS, SPFS and stablecoins—whereas subsea cables have few practical substitutes. A replacement cable cannot be laid during a crisis, so a major disruption could simultaneously impair payments, markets, military command-and-control and basic communications.

⁵⁰ Ohlen, Elsa. 2026. "Europe Doesn't Realize How Dangerous It Is": Telecoms CEO Warns of U.S. Dominance in Satellites, AI." CNBC. May 21, 2026. <https://www.cnbc.com/2026/05/21/starlink-spacex-satellites-telecoms-europe-dangerous-bouygues-ceo.html>.

⁵¹ Ranjit, Roshan, Robert Grindle, and Keval Khiroya. 2026. "European Satellites: Levelling Up." Deutsche Bank Research.

⁵² Yu, Edison, and Laura Li. 2026. "The Final Frontier: Space 101." Deutsche Bank Research.



The most important actor to watch may not be a state but a company: SpaceX. Starlink has reshaped satellite communications on cost, scale and performance, compressing the traditional hierarchy of space connectivity. Its ambitions now extend into markets historically served by terrestrial telecom and cable providers, in both advanced and emerging economies.⁵³ The implications for governments, defence and incumbent telecom operators will be significant.

In the era of infrastructure realism, power lies less in the number of cables or satellites built than in control of the network's junction points and architecture—from the ocean floor to low-Earth orbit. As with dollar-based payments and Taiwan's semiconductor capacity, concentration in cable landing rights and orbital slots creates a new class of systemic risk. The central question is no longer who wins the cable race, but who controls the rails through which global value flows—and on whose terms.

The final part of our series turns to data centres, moving the chokepoint further up the technology stack—from minerals and cables at the physical layer, through chips at the fabrication layer, to the infrastructure that stores and processes data at scale.

We would like to thank Edison Yu, Laura Li, Benjamin Black, Raymond Wong and Yi Xiong for their valuable insights and comments to this paper.

⁵³ Yu, Edison, Bryan Kraft, Ross Seymore, Benjamin Black, Laura Li. 2026. "SpaceX: Apex of Civilization Ambition." Deutsche Bank Research.



Appendix 1

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